Ex.No.1	**Study of various Network commands used in Linux and Windows:**
Date:	**Hands-on practice of various network commands.**

AIM: - To Study of various Network commands used in Linux and Windows

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	arp -a	arp -a	Displays the ARP cache, showing the IP addresses and corresponding MAC addresses of devices on the local network.	arp -a	Interface: 192.168.1.10 --- 0x7 Internet Address Physical Address Type 192.168.1.1 00-1a-2b-3c-4d-5e dynamic
2	hostname	hostname	Displays the hostname (computer name) of the local system.	hostname	DESKTOP-PC01
3	ipconfig /all	ipconfig /all	Displays complete TCP/IP configuration, including IP address, subnet mask, gateway, DNS, DHCP, and MAC address.	ipconfig /all	IPv4 Address : 192.168.1.10 Default Gateway : 192.168.1.1 DNS Server : 8.8.8.8 DHCP Enabled : Yes
4	nbtstat -a	nbtstat -a <hostname>	Displays NetBIOS name table of a remote computer for	nbtstat -a DESKTOP-PC01	NetBIOS Remote Machine Name Table Name Type Status DESKTOP-

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			troubleshooting NetBIOS over TCP/IP.		PC01 <00> UNIQUE Registered
5	netstat	netstat	Displays active TCP/IP connections, listening ports, routing tables, and interface statistics.	netstat	Proto Local Address Foreign Address State TCP 192.168.1.10:49762 142.250.190.78:443 ESTABLISHED
6	netstat -r	netstat -r	Displays the IP routing table of the local computer.	netstat -r	IPv4 Route Table Network Destination Netmask Gateway Interface
7	nslookup	nslookup <domain>	Queries the DNS server to obtain the IP address or DNS records of a domain.	nslookup www.google.com	Server: dns.google Address: 8.8.8.8 Name: www.google.com Address: 142.250.190.68
8	pathping	pathping <hostname/IP>	Combines the features of ping and tracert to trace the route and measure packet loss at each hop.	pathping www.google.com	Tracing route to www.google.com Computing statistics... Hop 1 192.168.1.1 Hop 2 10.0.0.1 Packet Loss: 0%
9	ping	ping <hostname/IP>	Tests connectivity between the local computer and another host using ICMP Echo	ping www.facebook.com	Pinging www.facebook.com [157.240.22.35] with 32 bytes of data: Reply from 157.240.22.35:

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			Request and Reply messages.		bytes=32 time=25ms TTL=54
10	route	route print	Displays or modifies the IP routing table. It is commonly used to view or configure static routes.	route print	IPv4 Route Table Active Routes: Network Destination Netmask Gateway Interface

Some important Linux networking commands

S.No	Command	Synopsis	Description	Sample Input	Sample Output
1	ip	ip <OPTIONS> <OBJECT> <COMMAND>	Displays and manages IP addresses, routing tables, and network interfaces.	ip address show	2: eth0: <UP,BROADCAST,RUNNING> inet 192.168.1.100/24 brd 192.168.1.255 scope global eth0
2	ip address add	ip address add <IP>/<Prefix> dev <Interface>	Assigns an IP address to a network interface.	ip address add 192.168.1.254/24 dev eth0	Assigned 192.168.1.254/24 to eth0 successfully.

S.No	Command	Synopsis	Description	Sample Input	Sample Output
3	ip address del	ip address del <IP>/<Prefix> dev <Interface>	Removes an IP address from a network interface.	ip address del 192.168.1.254/24 dev eth0	Deleted 192.168.1.254/24 from eth0.
4	ip link set up	ip link set <Interface> up	Brings the network interface online.	ip link set eth0 up	Interface eth0 is UP.
5	ip link set down	ip link set <Interface> down	Brings the network interface offline.	ip link set eth0 down	Interface eth0 is DOWN.
6	ip link set promisc on	ip link set <Interface> promisc on	Enables promiscuous mode on the interface.	ip link set eth0 promisc on	Promiscuous mode enabled on eth0.
7	ip route add default	ip route add default via <Gateway> dev <Interface>	Adds the default gateway for all outgoing traffic.	ip route add default via 192.168.1.254 dev eth0	Default route added successfully.
8	ip route add	ip route add <Network> via <Gateway>	Adds a static route through a gateway.	ip route add 192.168.1.0/24 via 192.168.1.254	Route added successfully.
9	ip route add (Direct Route)	ip route add <Network> dev <Interface>	Adds a direct route through a	ip route add 192.168.1.0/24 dev eth0	Route added via eth0.

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			network interface.		
10	ip route delete	ip route delete <Network> via <Gateway>	Deletes a static route.	ip route delete 192.168.1.0/24 via 192.168.1.254	Route deleted successfully.
11	ip route get	ip route get <Destination IP>	Displays the route used to reach a destination.	ip route get 10.10.1.4	10.10.1.4 via 192.168.1.254 dev eth0 src 192.168.1.100
12	ifconfig	ifconfig	Displays and configures network interface parameters (deprecated in modern Linux).	ifconfig	eth0: flags=4163<UP,BROADCAST,RUNNING> inet 192.168.1.100 netmask 255.255.255.0
13	mtr	mtr <hostname/IP>	Combines Ping and Traceroute to analyze network paths.	mtr google.com	HOST: localhost Loss% Snt Last Avg Best Wrst StDev
14	mtr -g	mtr -g <hostname>	Displays numeric IP	mtr -g google.com	192.168.1.1 → 142.250.190.78

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			addresses instead of hostnames.		
15	mtr -b	mtr -b <hostname>	Displays both hostnames and IP addresses.	mtr -b google.com	router.local (192.168.1.1) google.com (142.250.190.78)
16	mtr -c	mtr -c <count> <hostname>	Sends a specified number of probes.	mtr -c 10 google.com	10 probes completed successfully.
17	tcpdump -D	tcpdump -D	Lists all available network interfaces.	tcpdump -D	1.eth0 2.lo 3.wlan0
18	tcpdump	tcpdump -i <Interface>	Captures packets on a network interface.	tcpdump -i eth0	Listening on eth0...
19	tcpdump -c	tcpdump -i <Interface> -c <count>	Captures a specified number of packets.	tcpdump -i eth0 -c 10	10 packets captured.
20	tcpdump host	tcpdump -i <Interface> host <IP>	Captures traffic to and from a specific host.	tcpdump -i eth0 host 8.8.8.8	Captured packets to/from 8.8.8.8.

S.No	Command	Synopsis	Description	Sample Input	Sample Output
21	tcpdump src host	tcpdump -i <Interface> src host <IP>	Captures packets originating from a specific host.	tcpdump -i eth0 src host 8.8.8.8	Source packets from 8.8.8.8 displayed.
22	tcpdump dst host	tcpdump -i <Interface> dst host <IP>	Captures packets destined for a specific host.	tcpdump -i eth0 dst host 8.8.8.8	Destination packets to 8.8.8.8 displayed.
23	tcpdump net	tcpdump -i <Interface> net <Network>/<Prefix>	Captures traffic from a specified network.	tcpdump -i eth0 net 10.1.0.0/24	Packets from network 10.1.0.0/24 displayed.
24	tcpdump port	tcpdump -i <Interface> port <Port>	Captures traffic on a specific port.	tcpdump -i eth0 port 53	DNS packets captured.
25	tcpdump host and port	tcpdump -i <Interface> host <IP> and port <Port>	Captures traffic for a specific host and port.	tcpdump -i eth0 host 8.8.8.8 and port 53	DNS traffic with host 8.8.8.8 displayed.
26	tcpdump port not	tcpdump -i <Interface> port not <Port1> and not <Port2>	Captures all traffic except the specified ports.	tcpdump -i eth0 port not 53 and not 25	Traffic excluding ports 53 and 25 captured.
27	ping	ping <hostname/IP>	Tests network connectivi	ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=10.5 ms

S.No	Command	Synopsis	Description	Sample Input	Sample Output
			ty using ICMP Echo Requests and Echo Replies.		
28	ping -c	ping -c <count> <hostname>	Sends a specified number of ICMP Echo Requests.	ping -c 10 google.com	10 packets transmitted, 10 received, 0% packet loss.

Configuring an Ethernet connection by using nmcli:

Step No.	Task	Command	Expected Result
1	List all Network Manager connection profiles	nmcli connection show	Displays all available network connection profiles with their **Name, UUID, Type, and Device**.
2	Create a new Ethernet connection profile It is specifically an **nmcli connection management command**. • **nmcli** → NetworkManager Command Line Interface	nmcli connection add con-name Internal-LAN ifname enp1s0 type ethernet	Creates a new Ethernet connection profile named **Internal-LAN**.

Step No.	Task	Command	Expected Result
	• **connection** → Manages network connection profiles. • **add** → Creates a new connection profile.		
3	Rename an existing connection profile (Optional)	nmcli connection modify "Wired connection 1" connection.id Internal-LAN	Renames the existing connection profile to **Internal-LAN**.
4	Display the current settings of the connection profile	nmcli connection show "Internal-LAN"	Displays interface name, autoconnect status, IPv4 and IPv6 configuration details.
5	Configure IPv4 using DHCP	nmcli connection modify "Internal-LAN" ipv4.method auto	Configures the interface to obtain the IPv4 address automatically using DHCP.
6	Configure a static IPv4 address	nmcli connection modify "Internal-LAN" ipv4.method manual ipv4.addresses 192.0.2.1/24 ipv4.gateway 192.0.2.254 ipv4.dns 192.0.2.200 ipv4.dns-search example.com	Assigns a static IPv4 address, gateway, DNS server, and DNS search domain.
7	Configure IPv6 using SLAAC	nmcli connection modify "Internal-LAN" ipv6.method auto	Enables automatic IPv6 configuration using Stateless Address Autoconfiguration (SLAAC).

Step No.	Task	Command	Expected Result
8	Configure a static IPv6 address	nmcli connection modify "Internal-LAN" ipv6.method manual ipv6.addresses 2001:db8:1::fffe/64 ipv6.gateway 2001:db8:1::ffee ipv6.dns 2001:db8:1::ffbb ipv6.dns-search example.com	Assigns a static IPv6 address, gateway, DNS server, and search domain.
9	Activate the connection profile	nmcli connection up Internal-LAN	Activates the **Internal-LAN** network connection profile.
10	Verify the IP address configuration	ip address show enp1s0	Displays the assigned IPv4 and IPv6 addresses of the **enp1s0** network interface.
11	Verify the IPv4 default gateway	ip route show default	Displays the configured IPv4 default gateway.
12	Verify the IPv6 default gateway	ip -6 route show default	Displays the configured IPv6 default gateway.
13	Verify the DNS configuration	cat /etc/resolv.conf	Displays the configured DNS servers and DNS search domain.
14	Test network connectivity	ping <host-name-or-IP-address>	Sends ICMP Echo Requests to verify connectivity with the specified host.

Sample Verification Outputs

Verification Command	Sample Output
ip address show enp1s0	inet 192.0.2.1/24 inet6 2001:db8:1::fffe/64
ip route show default	default via 192.0.2.254 dev enp1s0 proto static
ip -6 route show default	default via 2001:db8:1::ffee dev enp1s0 proto static
cat /etc/resolv.conf	search example.com nameserver 192.0.2.200 nameserver 2001:db8:1::ffbb
ping google.com	64 bytes from google.com: icmp_seq=1 ttl=117 time=11.2 ms

OUTPUTS:

1.hostname

```
PS C:\Users\DELL> hostname
HDC0124094
```

2.arp -a

```
Trace complete.
PS C:\Users\DELL> arp -a

Interface: 192.168.56.1 --- 0x10
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.77.77.77          01-00-5e-4d-4d-4d     static
  225.0.48.21           01-00-5e-00-30-15     static
```

3.nslookup

```
PS C:\Users\DELL> nslookup google.com
Server:  UnKnown
Address:  172.16.72.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:805::200e
          142.250.67.46
```

4.ifconfig

```
student@fedora:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::55f0:9771:cb55:7bf3  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:2d:91:bf  txqueuelen 1000  (Ethernet)
        RX packets 64  bytes 23284 (22.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 115  bytes 13057 (12.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

5. ip address show

```
student@fedora:~$ ip address show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:2d:91:bf brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86231sec preferred_lft 86231sec
    inet6 fd17:625c:f037:2:eebe:9b47:404f:31ac/64 scope global dynamic noprefixroute
       valid_lft 86234sec preferred_lft 14234sec
    inet6 fe80::55f0:9771:cb55:7bf3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

RESULT:

Thus the various network commands used in Linux and Windows were studied and their outputs were observed successfully.